

Exam Date & Time: 14-May-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2025

AUTONOMOUS MOBILITY [COE-3201]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Differentiate between traditional and autonomous mobility systems (3)
 - A)
 - B) Briefly describe the role of each component: Perception, Decision Making, Control, and Communication in Autonomous Vehicles (4)
 - C) Describe applications of autonomous mobility in transport and delivery industry (3)
- 2) Briefly describe the working of LiDAR and how it helps in perception (4)
 - A)
 - B) What is the role of GPS and IMU in localization? (3)
 - C) Explain the role of AI in perception (3)
- 3) What is the purpose of path planning in autonomous vehicles (3)
 - A)
 - B) Compare any two global path planning algorithms with examples (4)
 - C) Explain the role of actuators in vehicle control with reference to throttle and steering. (3)
- 4) Explain the role of ROS/ROS2 middleware in autonomous vehicles. (4)
 - A)
 - B) State any three cybersecurity threats in V2X communication. (3)
 - C) What are Over-the-Air (OTA) updates and why are they important? (3)
- 5) Compare Level 2 and Level 4 of SAE autonomy (3)

- A)
- B) What is HIL, and VIL in the context of validation? Give one line on each. (4)
- C) Mention the level and two differences between Tesla Autopilot and Waymo's autonomous system (3)

-----End-----